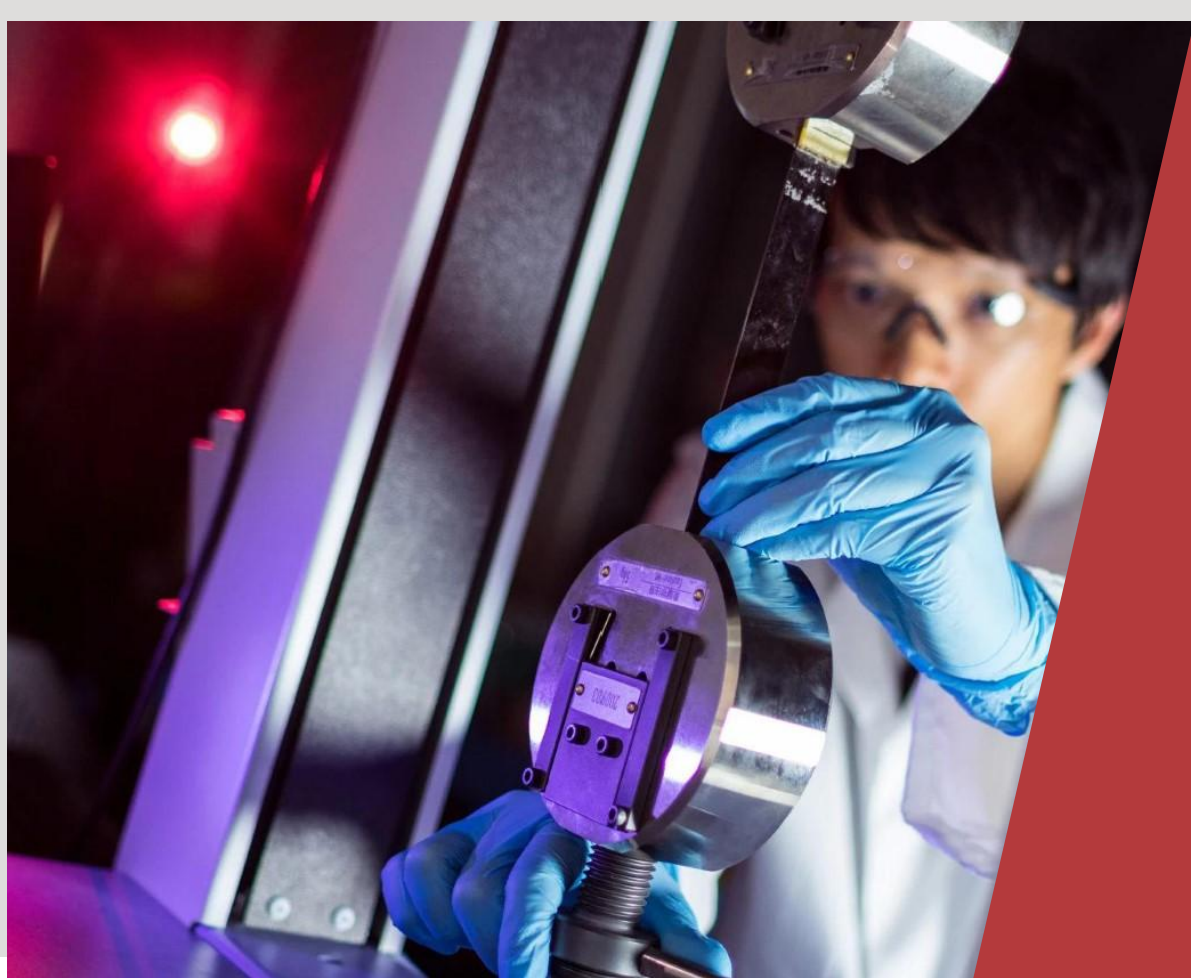




Signals and Dynamics Lab Report

Abstract

This experiment aimed to calculate the Young's modulus of a material via analysis of its resonant frequency. Due to the fundamental importance of the Young's modulus in material behaviour, it is of particular importance to obtain an accurate estimation of it when relying on a material's resistance to strain. The experiment involved using a microphone integrated into an Operational Amplifier circuit to calculate the period of an oscillating metallic specimen at varying free lengths. The method obtained a value for the Young's modulus of $196.2 \pm 33.2 \text{ GPa}$, which is consistent with the expected range of 190-200 GPa for stainless steel. This therefore suggests that this process has potential applications in situations requiring an inexpensive and quick way to analyse material property, where exact knowledge of the Young's modulus of the material is not essential.

1. Introduction

The Young's modulus is an intrinsic property that describes a material's resistance to deformation when under load, specifically relating material stress and resultant strain (British Engineering Services, 2025). The Young's modulus of a material has substantial implications regarding its behaviour under load, and therefore fundamentally dictates what applications it would be best suited for; For example, materials possessing a large Young's modulus, such as metals, are better suited for load-bearing structures (i.e. buildings, aircraft structures), compared to materials like polymers, which usually have a far smaller Young's modulus, and therefore deform more easily under stress.

Due to the ubiquity of the Young's modulus in material behaviour, it is possible to calculate it by methods other than via direct analysis (commonly by uniaxial tensile test). Since the Young's modulus affects the resonant frequency of a material, it is theoretically possible to calculate the Young's modulus of a material from its vibrational behaviour, analysing the resultant sound profile to estimate its Young's modulus.

This experiment aims to determine the reliability of this method of analysis, using an operational amplifier (op-amp) circuit to calculate a selected material's resonant frequency, and using the obtained results to calculate an estimated value for the tested specimen's Young's modulus.

2. Methods

In order to conduct the experiments, a microphone with high-pass filter was integrated into a op-amp circuit, with the primary sections dedicated to the microphone and its power supply, a DC offset trimmer, a differential amplifier, and a Raspberry Pi Pico for data collection, as depicted in Figure 1.

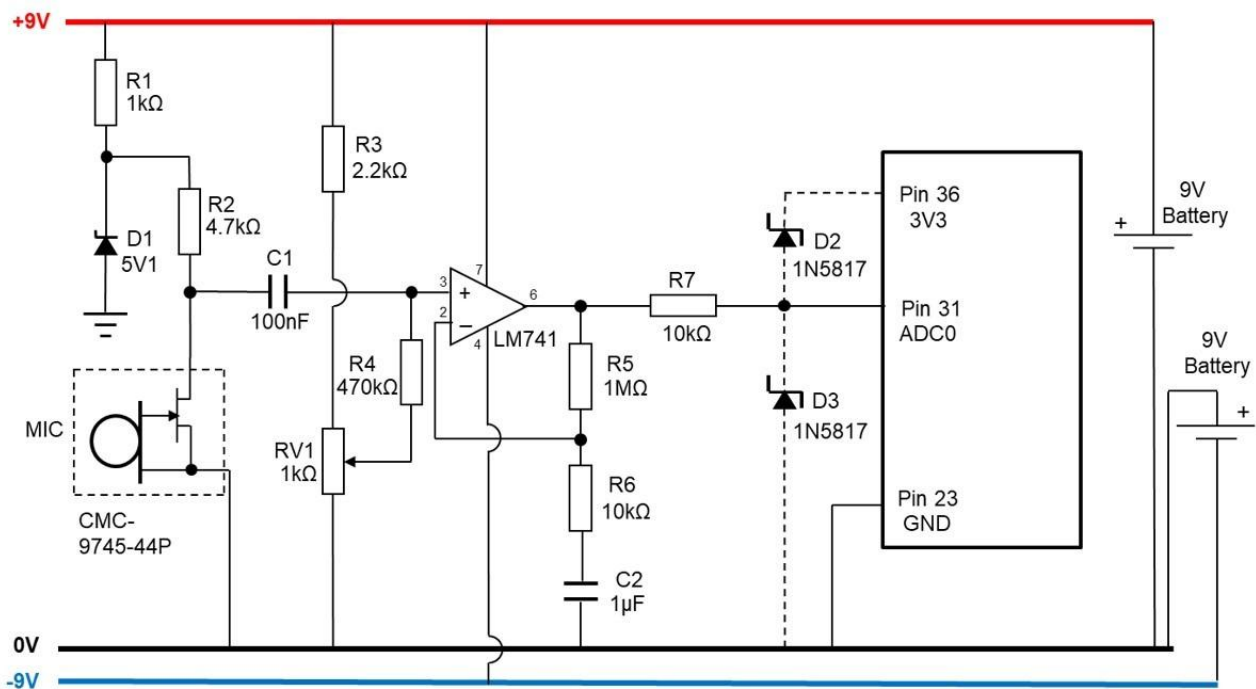


Figure 1 - Diagram of the microphone circuit

The circuit design allowed for an Analogue-to-Digital (ADC) conversion of the audio signal produced by the test specimen, with the digital output enabling analysis through use of suitable code. The circuit was constructed, as depicted in Figure 2, with the Pico connected to a computer to allow for data transfer and storage.

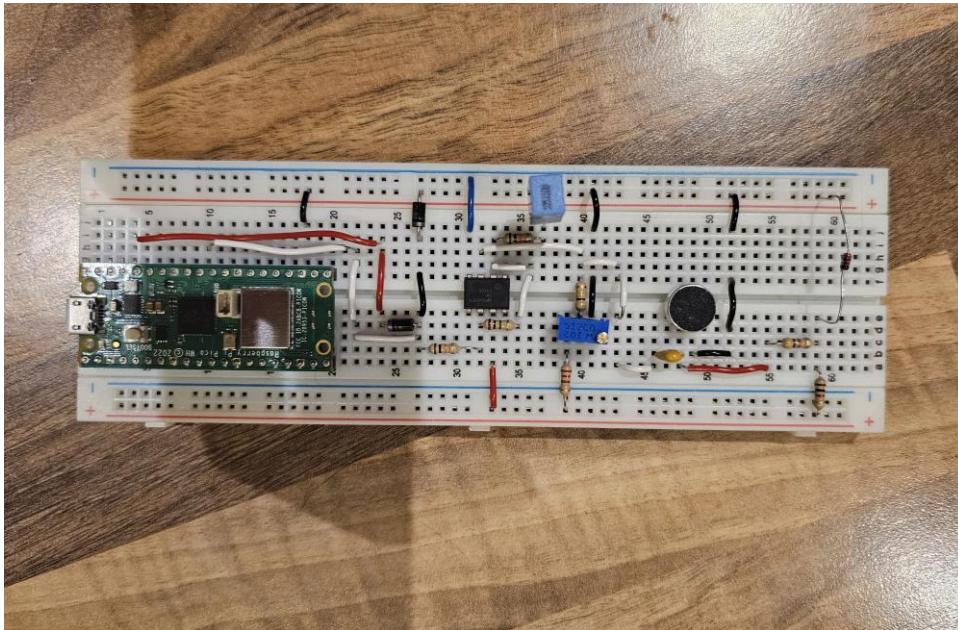


Figure 2 - Image of the completed circuit

The tested specimen was a 300mm metallic ruler. It was placed on the edge of a table with a fixed length overhanging the edge of the table, with the microphone placed 7cm directly below it, as shown in Figure 3, which was close enough to ensure the microphone received a clear sound without risk of the ruler colliding with the circuit. The non-overhanging end of the ruler was pinned to the table using a weighted item, as depicted in Figure 4, to ensure that only the overhanging portion of the ruler would be oscillating and it would not redistribute its bending moment further down its span. This therefore approximated a cantilever beam condition.

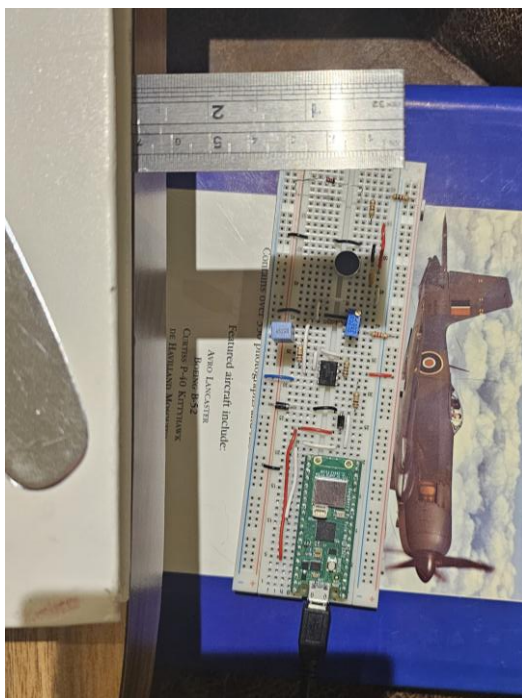


Figure 3 - Proximity of circuit board and oscillator



Figure 4 - Method of securing non-overhanging portion of oscillator to table

The selected values for analysis started from 6cm, and increased in 1cm intervals until 12cm. These values provided a good range to increase reliability whilst also ensuring the frequency of the pendulum would not be too high or low to reliably record the produced frequency. Each value was sampled five times to ensure repeatability and mitigate the effect of any anomalous readings.

The recorded audio, once converted into a digital signal by the circuit, was then graphed against the time history of the signal. Python code was developed allowing each unique point where the graph crossed the x-axis to be noted. The code cropped down the signal to a suitable range for analysis, where the average interval between half-crossings was calculated, in order to estimate the average half-period. This was ultimately converted into a value for frequency.

The uncertainty in the results was also considered; By calculating the percentage uncertainty for each value of overhang length, it was possible to plot maximum and minimum values for the Young's modulus, as well as a margin of error.

3. Results

According to theory, a direct relationship exists between the stiffness of an oscillator and its natural frequency (Blake, n.d.), with this relationship being described in equation [1] as below:

$$f = 0.56 \sqrt{\frac{EI}{qL^4}} \quad [1]$$

Where f is the fundamental frequency of the oscillator in Hertz (Hz), E is the Young's modulus of the oscillator in Pascals (Pa), I is the second moment of area in metres to the fourth power (m^4), q is the mass per unit length of the oscillator in kilograms per metre ($kg\ m^{-1}$), and L is the free overhang length of the cantilever in metres (m).

Rearranging this equation allows for a graph to be plotted of Length squared against the vibrational period, where the gradient of the graph will correspond to some constant relating to the Young's Modulus of the oscillator.

The obtained results are detailed in Table 1 - Results obtained from the experimentTable 1 below:

Table 1 - Results obtained from the experiment

Length [cm]	Length ² [cm ²]	Frequency [Hz]						Period (ms)
		Test 1	Test 2	Test 3	Test 4	Test 5	Mean	
6.0	36.0	162.3	159.8	168.2	168.8	159.0	163.6	6.1
7.0	49.0	117.1	119.0	112.9	119.0	119.3	117.5	8.5
8.0	64.0	100.0	97.9	105.9	98.3	98.9	100.2	10.0
9.0	81.0	71.4	77.8	80.1	80.1	77.4	77.4	12.9
10.0	100.0	58.5	56.4	63.8	58.5	58.0	59.1	17.0
11.0	121.0	47.9	48.8	49.9	49.9	49.5	49.2	20.3
12.0	144.0	47.8	41.5	40.0	47.8	46.8	44.8	22.4

The data was obtained by graphically analysing each dataset individually, as depicted in Figure 5.

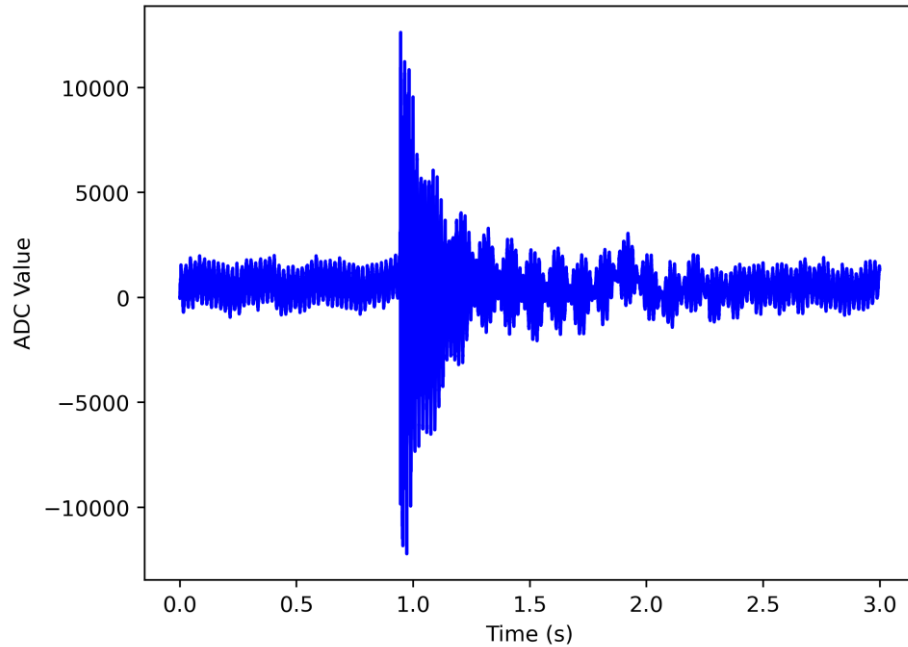


Figure 5 - Graph of Raw ADC Output

Each dataset was then further analysed by the code, with points of x-axis crossings logged and clearly outlined, as seen in Figure 6.

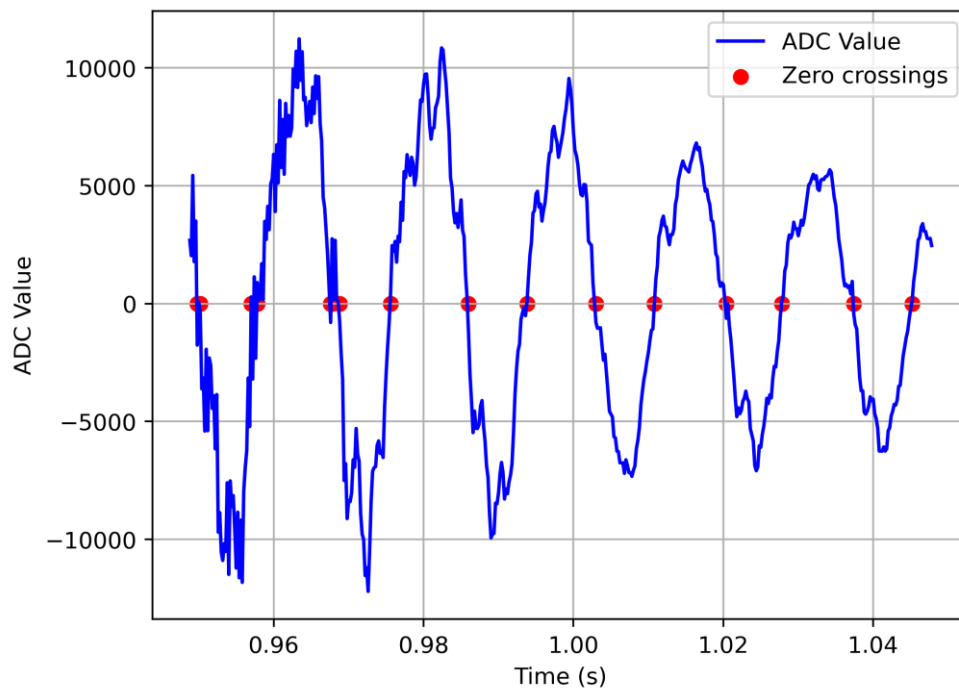


Figure 6 - Graph of cropped data, with zero crossings outlined

From this point, the resonant frequency of each test was calculated, with the average frequency for each length being calculated. At this point, the percentage uncertainty was also found for each length. This information was then collated onto one individual graph, where the obtained period values were plotted against the length squared. Linear regression was performed to determine a line

of best fit, as well as lines of maximum and minimum gradient, due to the error bars, as depicted in Figure 7.

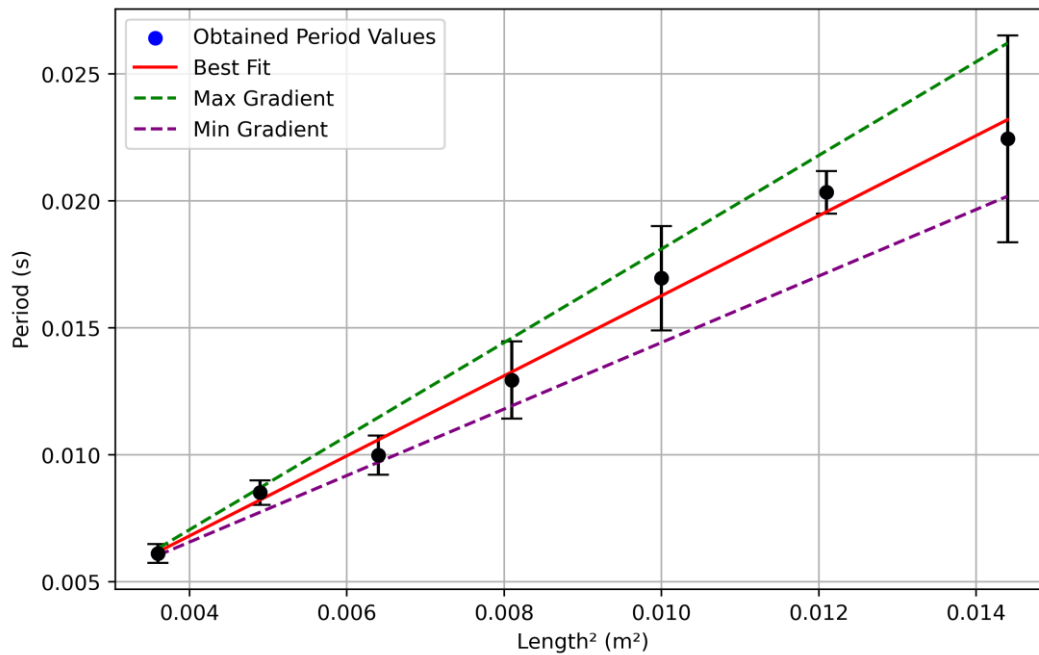


Figure 7 - Graph of obtained Period values against Length Squared

By rearranging equation [1], it was possible to obtain a value for the Young's modulus as a function of the gradient, which was evaluated to be approximately 196.2 ± 33.2 GPa.

4. Discussion

The experiment allowed for the estimation of the Young's modulus of a metallic specimen, obtaining an estimated value of 196.2 ± 33.2 GPa. Given that the actual Young's modulus of stainless steel (the primary material in the specimen) typically ranges between 190 to 200 GPa (Beam Dimensions, 2024), it can be said that this method of determining the Young's modulus is somewhat effective.

However, it should be noted that this obtained value has a very substantial range of error. This is likely due to a number of sources that were identified, including inaccuracies in measuring the exact free length of the oscillator. Since the obtained value for frequency is dependent on the length squared, then any inaccuracy in the measured oscillating length will result in an amplified margin of error. However, since the overall margin of error from this source is very low (± 1 mm), it is unlikely that this was the largest overall source of error. This source of error could be further reduced by taking more samples at each length (up to 10), and excluding anomalous values from analysis.

A further source of error was identified within the code used to analyse the frequency of the oscillating specimen. Due to the obtained ADC input not being a perfect sine wave, if the graph crossed the x-axis multiple times in a period of time far shorter than the actual oscillating period of the specimen, then each point where the graph crossed the x-axis would be separately labelled as its own individual crossing points, instead of being treated as one individual crossing point, as seen in the second and third labelled crossing points in Figure 6. Although the code was adapted to exclude crossing points

that occurred within 1 millisecond of another one¹, it is still possible that some anomalously labelled crossing times did not fall within this timeframe and therefore affected the results. This would have led to a positive systemic error in the calculated frequency, since more crossing points would have been recorded in the same time period. Since each sample of the recording used for analysis contained around 15-20 crossing points, an additional erroneous crossing point would increase the calculated frequency by approximately 6%, which would substantially affect the final results. This source of error could be reduced by adjusting the maximum time difference between two crossing points, to ensure all anomalous crossing points are disregarded, whilst making sure that two genuine independent crossing points are not classified as a single one. This could be found by calculating the expected frequency range of the specimen, which would provide a theoretical expected minimum time limit between crossing points.

Finally, it should be noted that the calculated percentage uncertainty is likely inaccurate. As seen in Figure 7, the maximum and minimum gradient lines do not pass through all of the error bars, meaning that a far lower percentage uncertainty value would have been calculated had the maximum and minimum lines of best fit gone through every error bar. This has a significant impact on the confidence of this calculated error range, suggesting that this method is more accurate than the error range suggests. This could be mitigated in future by adjusting the python code used to plot the figures to ensure that the lines of best fit passed through every error bar.

5. Conclusion

This experiment involved recording an oscillating metallic specimen to calculate its natural frequency, and therefore its Young's modulus. This method provided an estimate for the Young's modulus of the specimen as $196.2 \pm 33.2 \text{ GPa}$, which falls in the approximate range for stainless steel of 190-200 GPa, although this is dependent on the accuracy of the calculated percentage error. This has applications in areas requiring a basic evaluation of the Young's modulus of a material without access to high-accuracy tensile test machines, and where high levels of precision are not required. Further research should aim to reduce the effect of the sources of error highlighted in this report, through scrutinising the code used to find the maximum and minimum gradients, as well as calculating the expected frequency range of the specimen to prevent the counting of erroneous double crossings. Correcting these sources of error could allow for analysis into the maximum possible accuracy for finding a material's Young's modulus via this method.

6. References

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¹ i.e. so that if two or more crossings occurred in quick succession, then only the first one would be used for calculations, and the rest would not be included.

